

Shyamoli Engineering College
Dept. of Computer Science and Engineering
1st Year, 2nd Semester, 2nd Assessment Examination
Course Title: Fundamentals of Programming
Course Code: CSE-1201

Answer any **four** from the following Questions including No. 06

Full Mark-20

Time- 1 Hour

1. Explain the use of pointers in call by reference parameter passing for functions. 5
2. Write a C program using pointer to swap the values of two integer numbers. 5
3. What are the purposes of memory management function malloc () and calloc ()?. 5
4. Explain about the following: (i) String and character array. 5
5. Explain the concept of pointers arithmetic operations. 5
6. How we declare and initialize string variables. Write a program to find the length of a given string. 5

Shyamoli Engineering College
Dept. of Computer Science and Engineering
1st Year, 2nd Semester, 2nd Assessment Examination
Course Title: Digital Logic Design
Course Code: CSE-1202

Answer any **four** from the following Questions

Full Mark-20

Time- 1 Hour

1. What does the acronym stands for RAM, ROM & EPROM? 5
Draw the schematic diagram of (4×4) RAM memory.
2. Simplify the following function using Quine-McCluskey 5
method :-
 $f(A, B, C, D) = \sum m(1, 3, 5, 8, 9, 11, 15) + d(2, 13)$
3. Describe the working of J-K flip-flop and state the race around 5
condition.
4. Describe the working of R-S Latch using NAND gates only and 5
state the race around condition.
5. Design the function $F(A, B, C, D) = \sum m(1, 4, 5, 8, 10, 12, 13)$ 5
using 8x1 multiplexer
6. Construct a 1-to-32 demultiplexer using only two types of 5
demultiplexers: 1-to-4 and 1-to-8.

Shyamoli Engineering College
2nd assessment Examination-2025
Sub: Physics; PHY-1203
Department: CSE
1ST YEAR 2ND SEMESTER

Time: 1 hour

Marks: 20

Answer all the questions

1. (a) Define Resolving Power of grating and Derive the equation. 08
(b) Write down the difference between Fraunhofer & Fresnel Diffraction. 02

2. (a) State & Derive Brewster's law. 06
(b) The critical angle for certain wavelength of light in the case of a piece of glass is 40° . Find the polarizing angle for glass. 04

Shyamoli Engineering College
Dept. of Computer Science & Engineering
Level-1, Term-2, 2nd Assessment Examination
Sub: Methods of Integration & Diff. (MATH-1204)

Full Mark-25

Time- 1 Hour

(Answer all the following Questions)

1.a) Solve $\frac{dy}{dx} = 1 - y$, $y(0) = 0$ using Euler's method. Find y at $x=0.1$ and $x=0.2$. Compare the result with exact result. (07)

b) A 30 volt electromotive force is applied to an LR series circuit in which the inductance is 0.2 henry and resistance is 50 ohms. Find the current $i(t)$ if $i(0)=0$. Determine the current after a long time. (08)

2. Define Bounded below & Bounded above sequence with example. Test the given series whether it's converge or not. (2+4+4)

i) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n+1} + \sqrt{n}}$

ii) $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2-1}$

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$\frac{1}{\sqrt{n+1} + \sqrt{n}}$

Shyamoli Textile Engineering College
Dept. of Computer Science & Engineering
Year-1, Semester-2, Second Class Assessment Examination
Course Title: Developing English language Skill
Course Code: ENG 1205

Time: 01 Hour

Total: 25

1. Write down the answer with appropriate preposition.

0.5x10= 05

Last Sunday, I went (1) my friends to a beautiful park located (2) the edge of the city. We started early (3) the morning to avoid the traffic. The weather was pleasant, and the sky was filled (4) soft white clouds. We sat (5) a large tree and unpacked our breakfast. While eating, we talked (6) our plans for the upcoming semester. Later, we walked (7) the lake and took some pictures. There were ducks swimming (8) the water, and children playing (9) the playground nearby. By noon, we decided to return home (10) bus, feeling relaxed and refreshed after a lovely morning outdoors.

2. Sentence Conversion according to the instruction given in bracket. 10

1. He worked hard to succeed. (*Complex sentence.*)
2. When he saw the police, he ran away. (*Simple sentence.*)
3. He was poor but he was honest. (*Complex sentence.*)
4. Although he is rich, he is not happy. (*Compound sentence.*)
5. She is a good singer. (*Interrogative sentence.*)
6. Is this your pen? (*Assertive sentence.*)
7. Please open the door. (*Assertive sentence.*)
8. You should be kind to others. (*Imperative sentence.*)
9. It is a very beautiful morning. (*Exclamatory sentence.*)
10. I wish you success in life. (*Optative sentence.*)

3. Imagine you are applying for the position of Junior Electrical Engineer at Techno Spark Ltd., a reputed electronics company in Dhaka. Prepare a CV (Curriculum Vitae) with an application and send it to the HR of this company.

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